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PROFILE SUMMARY

Software Engineer with **4+ years** building distributed systems on **GCP** using **Go** and **Java**. I build production systems end-to-end: service design, APIs, infra, observability, CI/CD, and operational debugging. Strong focus on low-latency services, fault tolerance, and infrastructure reliability. Improved cross-service latency by **~30ms** after moving internal traffic from **REST** to **gRPC**. Cut deployment time by **60%** through Docker-based **CI/CD** pipelines. Built LLM infrastructure in Go with routing, retries, caching, tenant budgets, failover handling, and async execution across providers.

SKILLS

Languages: Go, Java, Python, SQL, HTML, CSS
Frameworks: Spring Boot, Spring MVC, Hibernate, gRPC, Microservices, Angular, Servlets, JSP
Databases: PostgreSQL, MongoDB, Redis, BigQuery, Elasticsearch
Cloud & DevOps: GCP (GKE, GAE, Spanner, Datastore), AWS, Kubernetes, Docker, Terraform, Jenkins, CI/CD, Git
Observability: Prometheus, Grafana, OpenTelemetry, Jaeger, Datadog, Kibana
Messaging: RabbitMQ, Google Pub/Sub, NATS
Security & APIs: OAuth 2.0, JWT, RESTful APIs, gRPC, Spring Security, Postman
Testing: JUnit, Spock, Sonar (80%+ coverage), JMeter, TDD
AI & Agentic: Google ADK, Gemini AI, OpenAI, RAG, MCP, Cursor, Copilot, LLM gateway design

WORK EXPERIENCE

Vendasta - Software Developer II

September 2024 - Current — *Chennai, India*

- Owned the inference routing service end-to-end. Evaluated REST vs gRPC for internal traffic, moved services to gRPC, and cut **p99** latency by **~30ms**.
- Built Go microservices on GKE with fault-tolerant gRPC communication. Increased system throughput by 40% while maintaining **99.9%** availability.
- Set up Docker-based **CI/CD** pipelines with Prometheus monitoring and zero-downtime deployments. Reduced deployment time by **60%**.
- Introduced TDD practices across 3 services and kept unit test coverage above **80%** using **JUnit** and **SonarQube**. Production bug volume dropped by **35%**.
- Ran design reviews, sprint planning, and PR reviews for the team. Mentored 2 junior engineers on debugging, testing, and service design.
- Added AI coding workflows using **Copilot** and **Cursor** for repetitive tasks, test scaffolding, and refactors.

Dataman Computer Systems - Associate Software Developer

February 2022 - September 2024 — *Kanpur, India*

- Built and shipped **5+** backend services handling **10k+** daily requests.
- Benchmarked **PostgreSQL** + **Redis** against a pure NoSQL setup before launch. Chose the relational approach and reduced average response time by **25%**.
- Integrated Cashfree payments and Twilio notifications with retry handling and exponential backoff. Increased payment success rate from **85%** to **98%**.
- Wrote internal design docs covering service boundaries, caching decisions, and data flow patterns. The team reused those patterns across later projects.
- Mentored junior developers on Hibernate ORM, debugging, clean architecture, and PR reviews.
- Worked closely with QA and product teams during sprint delivery. Kept the bug turnaround under 2 days.

PROJECTS

Inference Gateway

Technologies Used: *Go, gRPC, Protobuf, PostgreSQL, Redis, NATS, Docker, Prometheus, Grafana, Jaeger, OpenTelemetry, Ollama*

- Built a multi-tenant **LLM gateway** in Go routing traffic from the HTTP edge through a **gRPC** orchestration layer into upstream providers.
- Added API key authentication using **SHA-256** hashing and propagated tenant context through gRPC metadata.
- Built a routing engine that selected the cheapest model meeting each request's latency and cost limits. Added automatic fallback to local **Ollama** models when providers failed.
- Enforced tenant budgets and per-second rate limits using Redis. Returned **402** and **429** responses with Retry-After handling during overages.
- Added async execution over **NATS** with worker-based processing, job state tracking, and orphan job recovery after worker crashes.
- Reduced repeat-request latency through Redis response caching, idempotency replay keys with **24h** TTL, and read-through caching for tenant policy lookups.
- Added **jittered** retries, provider health checks, **Prometheus** metrics, and distributed tracing through **OpenTelemetry** and **Jaeger**.

Event Management Application

Technologies Used: *Java, Spring Boot, Hibernate, Spring MVC, Spring Security, PostgreSQL, RESTful APIs, Docker, Redis, RabbitMQ, HTML5, CSS3, Thymeleaf, Cashfree, Twilio*

- Built a full-stack event management platform handling scheduling, ticketing, attendee management, and payments.
- Split notifications into **RabbitMQ**-driven async workers to avoid blocking user-facing requests.
- Integrated Cashfree and Twilio with retry handling and delivery tracking.
- Applied **TDD** with JUnit and maintained quality gates through **SonarQube**.
- Added Redis caching during peak traffic and reduced database load by 40%.
- Automated deployments through Docker-based CI/CD pipelines and reduced release time by **60%**.

HACKATHON

Google AI Agent Development Kit Hackathon · *Software & AI Developer*

- Built a **multi-agent AI** system using **Google ADK** and Gemini models to process **100+** concurrent IoT sensor workflows.
- Added event-driven coordination between sensor, analysis, and action agents for low-latency decisions under unstable network conditions.
- Built **RAG** pipelines and lightweight **MCP-style** communication between distributed agents.
- Improved crop prediction accuracy by **35%** using ML forecasting pipelines.
- Reduced redundant **LLM** calls by improving orchestration and task delegation between agents.

EDUCATION

Dr. A.P.J. Abdul Kalam Technical University - Masters (Master Of Computer Application - MCA)

June 2020 - July 2022 *Kanpur, India*